

**REQUEST FOR PROPOSALS FOR
THE ESTABLISHMENT OF 250 MW / 1000 MWh
STANDALONE BATTERY ENERGY STORAGE SYSTEM
FROM 10 MW /40 MWh AC CAPACITY PROJECTS
ON BUILD, OWN AND OPERATE BASIS
WITH 15 YEAR OPERATIONAL PERIOD**

International Competitive Bidding (ICB)

**Tender No: TR/REP&PM/ICB/2026/001/C
July 03, 2026**

Information Copy Not for Bidding

National System Operator (Pvt.) Ltd.

No. 50, Sir Chittampalam A. Gardiner Mawatha, COLOMBO 00200, SRI LANKA



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Contents of RFP

Volume I – Instructions to Project Proponents

Volume II – Proposal Letters & Forms

Volume III – Model Energy Storage Agreement

National System Operator (Pvt.) Ltd.

No. 50, Sir Chittampalam A. Gardiner Mawatha, COLOMBO 00200, SRI LANKA



Definitions

1. "Request for Proposal" or "RFP" shall mean this Request for Proposal document comprising Volume I – Instructions to Project Proponents, Volume II – Proposal Letters & Forms, and Volume III – Model Energy Storage Agreement, together with all Annexes, Schedules, Exhibits, Addenda, amendments, clarifications, and other documents issued by NSO
2. "Project Proponent" shall have the meaning ascribed to it in Clause 5 of Volume I of RFP.
3. "Project Company" shall mean the successful Project Proponent who have submitted a Performance Security
4. "MW,AC " Unit of output power connected to the National grid.
5. "Proposal Security" means bid security

Acronyms

AC	Alternative Current
BESS	Battery Energy Storage System
CEB	Ceylon Electricity Board
COD	Commissioning Date
DL	Distribution Licensee
EIA	Environmental Impact Assessment
GOSL	Government of Sri Lanka
IEC	International Electrotechnical Commission
IEE	Initial Environmental Examination
IEEE	Institute of Electrical and Electronics Engineers
kWh	kilowatt-hour
MOP&E	Ministry of Power & Energy
NCRE	Non Conventional Renewable Energy
NSCC	National System Control Centre
NSO	National System Operator(Pvt) Ltd.
LKR	Sri Lankan Rupees
SCADA	Supervisory Control and Data Acquisition
SLSEA	Sri Lanka Sustainable Energy Authority
UL	Underwriters Laboratories

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VOLUME I

**INSTRUCTIONS TO PROJECT
PROPONENTS**

National System Operator (Pvt.) Ltd.
No. 50, Sir Chittampalam A. Gardiner Mawatha, COLOMBO 00200, SRI LANKA



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1 INFORMATION TO PROJECT PROPONENTS

1.1 Introduction to National System Operator (Pvt.) Ltd.

National System Operator Pvt Ltd (“NSO”) is the independent system operator established in Sri Lanka under the provisions of the Sri Lanka Electricity Act, No. 36 of 2024, as amended in 2025, enacted by the Parliament of Sri Lanka to implement reforms to the electricity industry and to strengthen the operational independence, transparency, and efficiency of the national power sector.

NSO is mandated to undertake the real-time operation, coordination, and management of the national electricity transmission system while ensuring the safe, secure, reliable, and economic operation of the national grid. The head office of NSO is located at No. 50, Sir Chittampalam A. Gardiner Mawatha, Colombo 02, Sri Lanka.

NSO is responsible for system balancing, generation dispatch coordination, grid security management, outage coordination, ancillary service management, renewable energy integration, and facilitating non-discriminatory access to the transmission network for generators, transmission and distribution licensees, and other grid-connected entities. In carrying out its functions, NSO operates in compliance with the Grid Code, applicable regulatory requirements, and internationally accepted utility practices.

As the system operator of Sri Lanka, NSO plays a critical role in enabling the transition towards a modern and sustainable electricity sector through the integration of renewable energy resources, battery energy storage systems (BESS), and advanced grid technologies.

The projects indicated in this Request for Proposals (RFP) are to be developed under a competitive bidding approach, and the successful project proponent(s) shall obtain all necessary statutory approvals prior to commencing the respective projects.

1.2 Scope of the Project

NSO on behalf of Cabinet Appointed Negotiation Committee (CANC) hereby invites Proposals in the prescribed format (“Proposal”) for the design, supply, delivery, erection, testing, commissioning, operation and maintenance of Standalone Battery Energy Storage system as described below, hereinafter referred to as the Facility, (on Built Operate and Own basis).

The Facility shall include (but not limited to) site, Batteries, Inverters, EMS, Power Park Controller, stand-by / auxiliary / emergency power requirements, water requirements, environmental impact mitigatory measures, transformers, switch gear & protection schemes, transmission lines (overhead or under-ground up to the Termination Point), SCADA facilities up to the Termination Point and all other appurtenant equipment to operate a Battery Energy Storage System. The facility must comply with the following order of precedence

- a) Functional Performance Requirements – Grid Integration (Annex A)
- b) Grid Code approved by Public Utilities Commission in Sri Lanka dated 22 July 2024 (Annex B)
- c) CEB Guide for Grid Interconnection of Embedded Generators (Annex C)

The primary purpose of each Facility is to provide energy shifting, enabling better utilization of renewable energy. They are expected to operate on dispatch instructions of National System Control Centre, hence proper communication establishment is essential for the projects. The contribution to frequency response during emergencies shall be a secondary objective of each Facility.

The total installed capacity of Battery Energy Storage system will be limited as stated in Clause 1.3 of volume I. The bidders will be selected through competitive bidding process based on variable price option for each Grid Substation.

A single Project Proponent shall be permitted to submit only one (1) Project Proposal for a given Grid Substation, except for the following Grid Substations:

- Mathugama
- Kurunegala
- Bolawatta
- Matara
- Galle
- Chunnakam

For the Grid Substations listed above, a single Project Proponent may submit a maximum of two (2) Project Proposals per Grid Substation.

Subject to the above limitation, a Project Proponent may submit different Project Proposals for multiple Grid Substations selected from the list specified in Clause 1.3 of Volume I. Accordingly, RFP documents shall be issued separately to each Project Proponent for each Proposal intended to be submitted, and each RFP document shall be issued upon payment of the applicable fee specified in the relevant newspaper advertisement.

The project proponent has to specify the grid substation to which the power plant is supposed to be connected (among the list in Clause 1.3 of Volume I) at the time of submitting the Proposal.

Project proponents have to submit proposals after carrying out feasibility studies considering all the project costs including transmitting power to the specified grid substation with proper communication facilities to enable integration with the National System Control Centre. The feasibility study report confirming that the proposed Battery Energy Storage System (BESS) to be installed at the selected project site is technically feasible to be connected to the designated Medium Voltage (MV) feeder of the distribution network.

It is the sole duty of the Project Company to obtain all approvals necessary to develop the standalone Battery energy storage system of the proposed project site. GOSL/NSO does not take any responsibility if the Project Company fails to obtain any of the mandatory approvals required for the project.

1.3 Grid Substation Location & Grid Connection Voltage Level

Identified Grid Substations for interconnection, the grid substation capacity limits for Battery Energy Storage systems to be connected, connection voltage level are given below:

	Grid Substation	Capacity Limit and Expected Storage for Grid Substation	GPS Coordinates of the Grid Substation	
			Latitude	Longitude
1	Ambalangoda	10 MW / 40 MWh (Beginning of life)	6°16'57.33"N	80° 4'27.18"E
2	Ampara	10 MW / 40 MWh (Beginning of life)	7°17'26.91"N	81°41'33.89"E
3	Bolawatta	10 MW / 40 MWh (Beginning of life)	7°17'36.10"N	79°51'57.28"E
4		10 MW / 40 MWh (Beginning of life)		
5	Chunnakam	10 MW / 40 MWh (Beginning of life)	9°44'28.70"N	80° 2'7.16"E
6		10 MW / 40 MWh (Beginning of life)		
7	Hambantota	10 MW / 40 MWh (Beginning of life)	6°11'6.04"N	81° 7'37.27"E
8	Horana	10 MW / 40 MWh (Beginning of life)	6°43'45.40"N	80° 8'25.81"E
9	Killinochchi	10 MW / 40 MWh (Beginning of life)	9°22'37.91"N	80°24'34.58"E
10	Kiribathkumbura	10 MW / 40 MWh (Beginning of life)	7°15'34.92"N	80°34'40.57"E
11	Kosgama	10 MW / 40 MWh (Beginning of life)	6°56'44.53"N	80° 8'24.05"E
12	Kurunegala	10 MW / 40 MWh (Beginning of life)	7°28'33.96"N	80°23'36.51"E
13		10 MW / 40 MWh (Beginning of life)		
14	Madampe	10 MW / 40 MWh (Beginning of life)	7°29'50.85"N	79°51'15.45"E
15	Mahiyanganaya	10 MW / 40 MWh (Beginning of life)	7°19'56.72"N	81° 1'27.00"E
16	Matara	10 MW / 40 MWh (Beginning of life)	5°57'53.30"N	80°32'5.79"E
17		10 MW / 40 MWh (Beginning of life)		
18	Mathugama	10 MW / 40 MWh (Beginning of life)	6°30'21.50"N	80° 3'37.72"E
19		10 MW / 40 MWh (Beginning of life)		
20	Monaragala	10 MW / 40 MWh (Beginning of life)	6°50'6.90"N	81°18'54.76"E
21	Galle	10 MW / 40 MWh (Beginning of life)	6° 2'44.50"N	80°14'45.83"E
22		10 MW / 40 MWh (Beginning of life)		

23	Panadura	10 MW / 40 MWh (Beginning of life)	6° 43' 30" N	79° 55' 51" E
24	Pannala	10 MW / 40 MWh (Beginning of life)	7°19'16.18"N	79°59'50.25"E
25	Pannipitiya	10 MW / 40 MWh (Beginning of life)	6°50'37.87"N	79°56'28.23"E

Connection Voltage Level : 33kV

Plant capacity and Bidding options for Bidders:

Subject to not exceeding the **maximum grid capacity limits** specified in the above table the Project Proponents can offer proposals as below,

- Single Project per grid substation within the grid capacity limit, except for Mathugama, Kurunegala, Bolawatta, Matara, Galle, and Chunnakam grid substations where a single Project Proponent may submit a maximum of two (2) Project Proposals.
- Capacity range 10 MW/ 40 MWh, AC for each offer with + 10% variance in capacity or energy storage capacity (MWh).
- Project Proponents shall purchase RFP's separately for each and every proposal intend to submit for each grid substation.

1.3.1 Land

Project Company shall purchase or lease the land of approximately 80 perches (0.5 acre) per Battery Energy Storage system Project. In the event a Project Company proposes to utilize a land extent less than 80 perches, the Bidder shall be required to provide a detailed technical justification demonstrating the adequacy of the reduced land extent. Such justification shall include, inter alia, a comprehensive site layout plan indicating the arrangement of all major components of the BESS Facility (including battery containers, power conversion system, transformer(s), switchgear, control building, access roads, fire safety zones, and maintenance clearances), together with all relevant dimensions, safety setbacks, and operational clearances in accordance with applicable standards and good engineering practice.

Geographically separated land lots are not allowed to bid for a single Battery Energy Storage system Project. It is required to submit one of the following set of documents to confirm the ownership or the ability to procure or lease the selected land.

- A copy of the Deed of the land in favour of the Project Proponent, certified by an Attorney-at-Law with the survey plan (**clearly indicating the land extent to be utilized for the plant**) if the project proponent is the current owner. It is compulsory to register the deed in the Land Registry to consider for evaluation. (You may submit supporting documents to claim the same)
- A copy of the Lease Agreement (at least for 17 years) of the land in favour of the Project Proponent, certified by an Attorney-at-Law with the survey plan (**clearly indicating the land extent to be utilized for the plant**) if the project proponent has already leased the land. It is compulsory to register the Lease Agreement in the Land Registry to consider for evaluation (You may submit supporting documents to claim the same)
- "Agreement to sell" in favour of the Project Proponent, attested by a Notary Public, **reserved for a period of at least 6 months** from the closing date of submission of proposals, (as

indicated in the given format of Section 10, Volume II) with the certified copy of the Deed of the land and the survey plan (**clearly indicating the land extent to be utilized for the plant**).

- “Agreement to lease” (at least for 17 years) in favour of the Project Proponent, attested by a Notary Public, **reserved for a period of at least 6 months** from the closing date of submission of proposals, with the certified copies of the Deed of the land and the survey plan (**clearly indicating the land extent to be utilized for the plant**).

Note: Where a conflict arises due to the selection of the same land plot for multiple projects connected to the following grid substations, priority for the use of such land shall be granted to the Project with the lowest evaluated price among the affected Projects. The Project Proponent of the remaining affected Project shall, upon notification by the Procuring Entity, identify and secure an alternative land plot that fully complies with all technical, environmental, legal, and commercial requirements set out in this RFP. Any alternative land plot proposed shall be subject to the review and acceptance of the Procuring Entity. All costs, risks, and obligations associated with securing the alternative land plot shall be borne solely by the relevant Project Proponent.

- Mathugama
 - Kurunegala
 - Bolawatta
 - Matara
 - Galle
 - Chunnakam
- In the case of LRC land (Land Reform Commission), approval letter issued by LRC indicating that land can be leased out (at least for 17 years) or sold for this BESS Project. The relevant survey plan (**clearly indicating the land extent to be utilized for the plant**) and complete detail of the land shall be provided with the proposal.
- In the case of land owned by a state institution, an approval letter from the Secretary of the relevant line Ministry or an authorized officer stating that the particular land could be provided (at least for 17 years) in the future to the project proponent for the establishment of the proposed BESS Project. The relevant survey plan (**clearly indicating the land extent to be utilized for the plant**) and the complete detail of the land shall be provided with the proposal.
- In the case of state land, an approval letter from the Secretary of Ministry of land or an authorized officer stating that the particular land could be provided (at least for 17 years) in the future to the project proponent for the establishment of the proposed BESS Project. The relevant survey plan (**clearly indicating the land extent to be utilized for the plant**) and the complete detail of the land shall be provided with the proposal.

It is also required to provide the location of the selected land as per the Section 13, Volume II of the RFP.

Option to Change the Land After Awarding Stage:

If the Project Company is unable to implement the Battery Energy Storage System Project in the land which is indicated in the RFP due to practical reasons beyond its control,

1. The Project Company shall immediately notify NSO with supporting documents for the claimed reasons. The selected new suitable land shall be able to connect to the same Grid Substation.
2. Based on the above, NSO at its sole discretion shall notify the approval of the land change.
3. Upon the receipt of NSO approval, the Project Company shall promptly act to obtain the deed or lease agreement as per above.
4. The risk with regard to the incremental cost of the new Grid Interconnection arrangement shall be absorbed within the originally offered proposal Capacity Charge Rate.

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1.4 Project Milestones Schedule

Activity	Date
Issue of RFP Documents (Date of Advertisement)	July 03, 2026
Pre-Bid Meeting	July 24, 2026 at 10.00hrs (21 Days from Advertisement)
Requests for Clarifications up to	July 21, 2026 (32 Days from Advertisement)
Closing date of submission of Proposals	August 14, 2026 by 10.00hrs (42 Days from Advertisement)
Issue of Letter of Award	September 29, 2026 (after Evaluation, CANC & Cabinet approval)
Acceptance of Letter of Award	October 06, 2026 (Within 7 days from the receipt of Letter of Award)
Signing of Energy Storage Agreement (ESA)	November 06, 2026 (Within 01 month from the date of acceptance of Letter of Award)
Submission of Models and results on Dynamic Model Tests	December 07, 2026 (Within 01 month from the date of signing of ESA)
Submission of Environmental Clearance	December 07, 2026 (Within 01 month from the date of signing of ESA)
Financial Closure	January 06, 2027 (Within 02 months from the date of signing of ESA)
Commissioning, Testing & Commercial Operation Date	May 06, 2027 (Within 06 months from the date of signing of ESA)

1.5 Disclaimer

- a. The content of this invitation is provided to Project Proponents to assist them in obtaining a general understanding of the proposed project. It does not constitute a recommendation to Project Proponent to participate in the proposed project.
- b. The information, estimates or opinions are based on present circumstances, intentions and beliefs and may require subsequent modification. While the NSO has taken all reasonable care to ensure that the information in this RFP is accurate, they make no representation or warranty, expressed or implied, nor takes any responsibility of any kind with respect to the completeness or accuracy of any of the information contained herein. Therefore, the NSO will not be liable for any loss or damage that may arise from interpretations, errors or omissions from this RFP.
- c. Project Proponents should not rely on presentation made by government employees or their agents in relation to this project, other than expressly provided for herein.
- d. Project Proponents shall bear all costs and expenses incurred by them associated with the preparation and submission of their Proposal. The NSO shall not be responsible or liable for such costs and expenses, including but not limited to those of professional advice. Project Proponents shall have no rights to claim costs specified above.

- e. NSO shall have the right to accept or reject any or all the Proposals received. NSO reserves the right to terminate the process after the receipt of Proposals. NSO reserves the right to terminate the process if adequate competition has not been created or/ and the Capacity Charge Rates quoted by project proponents are not acceptable to the NSO.
- f. There shall be no contractual or other obligations by the NSO arising from this RFP.
- g. NSO shall have the right to seek any further information and/or clarifications that they may require from Project Proponents.
- h. NSO reserves the right not to disclose any details regarding the evaluation process.
- i. No verbal agreement or conversation with any officers, agents or employees of the GOSL/NSO, either before or after the execution of the Project
- j. NSO shall not be responsible for any interpretations or conclusions by the Project Proponent based on data furnished by the NSO or which the Project Proponent may obtain or arrive at from information given in the RFP.

2 INSTRUCTIONS TO PROJECT PROPONENTS

2.1 Visits to Grid Substation

Project Proponents may visit the relevant Grid Substation to obtain an understanding on the interconnection arrangement (with prior arrangement with the NSO/NTNSP), 30 days prior to the closing date of submission of proposals.

2.3 Amendments to the RFP

The NSO reserves the right to amend, revise or modify this RFP. Any amendments, revisions or modifications of this RFP shall be made through the issuance of Addenda by the NSO and a copy of each Addendum shall be sent to all Project Proponents (who have purchased the RFP document as at the date of issuance of Addenda), on the same day by same means. **The RFP may be so amended, revised or modified by way of such Addendum up to 7 days before the closing date for the submission of Proposals.** However, it shall be the responsibility of the Project Proponents to ensure that they have obtained all such Addenda from the NSO.

If any Addendum is of a nature, which may require substantial changes in the proposals, the closing date may be extended by a number of days as in the opinion of the NSO, such Addendum shall enable Project Proponents to reconsider or revise their Proposals.

In case where NSO decide to extend the Closing Date for Submitting Proposals, it will be published in Newspapers and NSO website. **The validity of the proposal and proposal security will be as per the extended closing date.**

2.4 Requests for Clarifications

To be eligible to request clarifications it is compulsory to purchase a RFP at the time of requesting the clarifications. Project Proponents seeking any clarifications regarding this RFP may submit their written requests by facsimile, or by registered mail to:

The Chairman CANC
C/o; Director (RE & ESCP)
National System Operator (Pvt) Ltd.
Meethotamulla Road,
Kolonnawa 10600,
SRI LANKA
FAX: + 94 11 2697022

Such requests should be received no later than the date specified in **Project Milestones Schedule**, Clause 1.4, Volume I of RFP in order to be discussed at the pre-bid meeting. NSO will not entertain any request for clarifications of the RFP after the clarification deadline.

2.5 Pre-Bid Meeting

A Pre-Bid Meeting will be held as per Project Milestones Schedule in Clause 1.4, Volume I of RFP at the,

**Office of the Director (Renewable Energy & Energy Storage Capacity Procurement)
National System Operator (Pvt.) Ltd.**
Meethotamulla Road,
Kolonnawa 10600, SRI LANKA.

Note: This venue is subjected to change based on the responses to the tender.

The Clarifications provided at the Pre-Bid Meeting shall strictly relate to any explanations that may be required in relation to the RFP documents issued to the prospective Project Proponents. The Pre-Bid Meeting is not intended to supplement or substitute the due diligence required to be undertaken by each Project Proponent. However, not attending to this Pre-Bid Meeting shall not disqualify a Project Proponent from furnishing a Proposal. Only those who have purchased this RFP or their authorised representatives shall be permitted to attend this meeting **(only two participants are allowed per Project Proponent)**. A copy of the Paying-in-Voucher (PIV) issued by NSO for the purchase of the RFP document is required to be produced at the entrance.

2.6 Preparation of the Proposal

In order to prove compliance with all the requirements of this document, the project proponent **shall submit the duly filled Forms given in the Volume II of this document**, together with all other information/documentary proof requested in this RFP. The language of the proposal and all documents related to the proposal shall be in English language.

2.7 Technical and Commercial Data

The Technical and Commercial data /details furnished by the Project Proponent should demonstrate requisite knowledge of project requirements and understanding of the required tasks. The Proposal should be formulated to indicate the general approach or methodology that is proposed for the performance of Project including such detailed information that is considered relevant. **The requested information as detailed out in Clause 3 shall be forwarded along with the Proposal.**

The Project Proponent is also required to forward its background, organisation of its consortium partners in the prescribed format.

2.7.1 Documents required to be submitted with the RFP

A Project Proponent shall submit a Proposal accompanied with the following documents. The submitted documents should be clearly tagged with the item number specified in the list below:

- a. Duly completed Proposal Letters and Forms including documents to be submitted as given in Volume II of RFP document.

- b. The requested information as detailed out in Clause 3, Volume I of RFP, shall be forwarded along with the Proposal.
- c. The feasibility study report as per Clause 2.7.2, Volume I of the RFP document. (EIA or IEE reports are not necessary).
- d. If the project proponent himself is the tenderer, it is mandatory for the Project Proponent to register under the Public Contract Act No. 3 of 1987 – (PCA3) of Sri Lanka. The original of the certificate issued by the registrar shall be submitted along with the Proposal or certified true copy by the Project Proponent.
- e. If the Project Proponent appoints a local agent, representative or nominee to act on behalf of the Project Proponent, such a local agent, representative or nominee shall register himself with under the Public Contract Act No. 3 of 1987 – (PCA3) of Sri Lanka. The original of the certificate issued by the registrar shall be submitted along with the Proposal or certified true copy by the Project Proponent.
- f. Duly executed Board Resolution or Power of Attorney to the authorised representative who is to sign on behalf of the Project Proponent.
- g. In case of the Project Proponent being a Joint Venture/consortium, the Joint venture/consortium agreement /letter of intent to enter into JV and /or license which define precisely the general conditions under which it will function, the sponsoring member authorised to represent it, the shares of each participating member and the particular obligation of each and every member under the joint venture/consortium, all of whom shall be jointly and severally liable for any or all obligations under the Project Documents with NSO.
- h. Proposal Security as specified in Clause 2.17.1, Volume I of the RFP.
- i. Any other documents the Project Proponent considers necessary to demonstrate its ability to perform the work to be contracted.
- j. Additional documents, if any, to support the proposal.

Note: The Project Proponent shall submit only one (1) Technical Proposal for each Proposal submitted under this RFP. The submitted Technical Proposal shall clearly define the proposed technical solution, configuration, equipment specifications, design approach, and all associated technical parameters relevant to the proposed Facility. Submission of multiple technical options, variant proposals, optional configurations, or alternative technical arrangements under the same Proposal shall not be permitted. Any such alternative Technical Proposals submitted in addition to the primary Technical Proposal shall not be considered during the evaluation process and shall be deemed non-responsive for evaluation purposes.

2.7.2 Feasibility Study Report

The feasibility study report shall be submitted considering all the project cost including transmitting power to the specified grid substation and infrastructure to communicate with the NSCC. The report

may include the project details, technical details as per Clause 3, Volume I including power transmission and financing plan as per Clause 2.8.2, Volume I of the RFP document.

2.7.3 Qualifications

Subcontractors or Manufacturers for the following major items of plant and services must meet the following minimum qualification criteria, herein listed for a Subcontractor for that item. List of projects/sales records certified by manufacturers should be submitted as evidence (As per Item 10 in Section 3 of Volume II) for the following minimum qualification criteria.

Item No.	Description of Item	Minimum Criteria to be met
1	Battery module & cells	Within the last three (3) years, manufacturer must demonstrate: At least 1 GWh of installed battery modules/ cells
2	Inverter	Within the last three (3) years, manufacturer must demonstrate: At least 1 GW of installed inverter products
3	Power Plant Controller (PPC) – this is the controller that integrates inverters over the SCADA to provide a coordinated response to grid conditions and dispatch instructions	Within the last three (3) years, manufacturer must demonstrate: At least two (2) projects using the same PPC product or product family as that offered for this project

2.8 Capacity Charge Rate Proposal

The Project Proponent shall submit its Capacity Charge Rate (excluding VAT) Proposal as per Section 4 of Volume II.

2.8.1 Capacity Charge Rate

The contract period is for 15 years. During this period, NSO undertakes to utilize the storage facility at the agreed capacity charge per month based on its availability. The capacity charge payable will be a flat Capacity Charge Rate for the entire contract period. There will not be any indexation to the Capacity Charge Rate during the entire Contract period.

One time Capacity Charge Rate Adjustment (Applicable Capacity Charge Rate) :

To consider the variation in foreign currency component involved in the Capital Cost of the Project and to absorb a portion of foreign exchange risk, the proposed Capacity Charge Rate will be re-adjusted once the incorporated to the Energy Storage Agreement as given in the following formula.

Capacity Charge Rate proposed by the Project Proponent = Y LKR/MW

85% of the proposed tariff will be re-adjusted based on the US Dollar (USD) variation:

Applicable Capacity Charge Rate: $0.15 \times Y + (0.85 \times Y \times P2/P1)$ LKR/kWh

Where,

- P1 - LKR/USD Indicative Exchange Rate applicable on the date of closing the proposals and published by the Central Bank of Sri Lanka
- P2-LKR/USD Indicative Exchange Rate applicable on (07) seven working days prior to the “date of signing the Energy Storage Agreement (D2)” and publish by Central Bank of Sri Lanka.
 - D2 (Date of signing the Energy Storage Agreement) – Applicable up to a maximum of 01 month from date of Letter of Award, in deriving P2
 - In unavoidable circumstances, where the ESA signed after 01 month from date of Letter of Award, with the consent of NSO, P2 shall be derived for the date that falls on (07) seven working days prior to the date” reaching 01 month from the date of Letter of Award”.

The project proponent shall make the BESS available for 400 full equivalent cycles per year with at least 20 cycles per month. Following provisions shall be applicable on the entire Project Capacity:

- i. The procurement shall be in power (MW) terms. The Project Company shall install, operate and maintain the BESS to offer facility to the National System Control Centre to charge and discharge the BESS on an “on demand” basis. The project proponent shall guarantee a **minimum system availability of 97%** on monthly basis. The project proponent shall pay the liquidated damages for such shortfall and shall duly pay such damages to NSO. Amount of such liquidated damages shall be twice the Capacity Charges for the capacity not made available.

However, at the conclusion of each contract year, if the average system availability over the year meets or exceeds the 97% threshold, the NSO shall carry out a reconciliation. Any liquidated damages imposed during that contract year for unavailability shall be reviewed, and necessary adjustments shall be made in favor of the Project Company, where applicable.

- ii. In addition, the Project Company shall also demonstrate, on annual basis, 100% of the minimum Dispatchable capacity of the BESS. Taking into consideration capacity degradation, the minimum dispatchable energy to be made available by the Project Company at the end of a given year shall be as follows: In the event of reduction of available storage capacity below the Minimum Dispatchable Capacity at the end each contract year, monthly capacity charge will be reduced based on the proportional loss of storage capacity.

Contract Year	Minimum Dispatchable Storage Capacity (MWh) at the end of Contract Year as a percentage of Contracted Storage Capacity (MWh)
1	97.5%
2	95.0%
3	92.5%
4	90.0%
5	88.0%
6	86.0%
7	84.0%
8	82.0%
9	80.0%
10	78.0%
11	76.0%
12	74.0%
13	72.0 %
14	70.0%
15	68.0%

iii. The Project proponent shall guarantee a **minimum AC to AC roundtrip efficiency (RtE) minimum 85%** for the system on monthly basis. The Project proponent shall be liable for Liquidated Damages, on account of conversion losses, based on the following conditions:

iv. For RtE <85%, there shall be a liquidated damage levied upon the excess conversion losses at 150% per unit of Peak time Tariff for supply of 33kV connected customers for General Purpose requirements. The tariff applicable would be rate applicable in each month during the operation of the facility.

v. For RtE ≥ 85%, there shall be no incentive payment or any liquidated damage levied for energy converted.

$$\text{Roundtrip Efficiency (RtE}_m\text{)} = \frac{\text{Total of actual energy discharged in the month}_m}{\text{Total of actual energy charged in the month}_m}$$

For detailed calculation of payments and penalties it is required to refer the Energy Storage Agreement in Volume III.

In the Proposal, the Project Proponent shall indicate the proposed flat Capacity Charge Rate for utilizing the storage facility based on dispatch instructions of National System Control Centre.

2.8.2 Financial Plan

The Project Proponent shall furnish a detailed financing plan with documentary proof to prove the financial soundness of the project proponent. The required information shall be provided in the Financial Evaluation Forms prescribed in Section 07 of Volume 02 and in accordance with the requirements specified in Tables 2.8.2.1 and 2.8.2.2.

2.8.2.1 Historical Financial Performance

Criteria	Compliance Requirements				Documents
Requirement	Single Entity	Joint Venture			Submission Requirements
		All Partners Combined	Each Partner	One Partner	
Submission of audited financial statements or, if not required by the law of the Bidder's country, other financial statements acceptable to the Employer, for the last three (03) years to demonstrate the current soundness of the Bidder's financial position. As a minimum, the Bidder's net worth for the last year calculated as the difference between total assets and total liabilities should be positive	Must meet requirement	Not applicable	Must meet requirement	Not applicable	Section 7 of Volume II - Form FIN - 1

2.8.2.2 Financial Resources

If the bid evaluation process and the decision for the award of the Contract takes more than 1 year from the date of bid submission, Bidders may be asked to resubmit their current contract commitments and latest information on financial resources supported by latest audited accounts or audited financial statements, or if not required by the law of the Bidder's country, other financial statements acceptable to the Employer, and the Bidders' financial capacity, will be reassessed on this basis.

Criteria	Compliance Requirements				Documents
Requirement	Single Entity	Joint Venture			Submission Requirements
		All Partners Combined	Each Partner	One Partner	
For Single Entities: The Bidder must demonstrate that its financial resources defined in FIN - 2, less its financial obligations for its current contract commitments defined in FIN - 3, meet or exceed	Must meet requirement	Not applicable	Not applicable	Not applicable	Section 7 of Volume II - Form FIN - 2 and Form FIN - 3

the total requirement for the Subject Contract of 600 million.					
For Joint Ventures: (1) One partner must demonstrate that its financial resources defined in FIN - 2, less its financial obligations for its own current contract commitments defined in FIN-3, meet or exceed its required share of 40% from the total requirement for the Subject Contract. AND	Not applicable	Not applicable	Not applicable	Must meet requirement	Section 7 of Volume II - Form FIN - 2 and Form FIN - 3
2).Each partner must demonstrate that its financial resources defined in FIN - 2, less its financial obligations for its own current contract commitments defined in FIN - 3, meet or exceed its required share of 25% from the total requirement for the Subject Contract. AND	Not applicable	Not applicable	Must meet requirement	Not applicable	Section 7 of Volume II - Form FIN - 2 and Form FIN - 3
(3) The Joint Venture must demonstrate that the combined financial resources of all partners defined in FIN - 2, less all the partners' total financial obligations for the current contract commitments defined in FIN - 3, meet or exceed the total requirement for the Subject Contract of Rs.600 million .	Not applicable	Must meet requirement	Not applicable	Not applicable	Section 7 of Volume II - Form FIN - 2 and Form FIN - 3

2.8.2.3 Estimated Project Cost

The project proponent also shall provide the detailed financing structure, including equity-debt split, cost of capital, loan terms, repayment profile, revenue assumptions, DSCR, and a full financial model with sensitivity analysis, key risk mitigating factors for the proposed 10 MW / 40 MWh BESS project.

The project proponent shall provide detailed project cost breakdown in accordance with formats specified in in Section 7 of Volume 2

In case if a single Project Proponent is bidding for more than one Grid Substation Debt, Equity eligibility criteria will be considered in total for all projects.

In case a single Project Proponent has applied for multiple projects at different Grid Substations, the Project Proponent must indicate each application by assessing ranking with 1 representing the

highest priority. The fulfilment of financial eligibility at final stage will be decided based on the provided priority.

2.8.3 Currencies of Capacity Charge Rate and Payments

All Capacity Charge payments shall only be in Sri Lanka Rupees. Payments for the purchase of storage services in respect of this Project would be calculated in Sri Lanka Rupees and will be paid in Sri Lanka Rupees. The Capacity Charge calculations shall not be based on any other currencies or any baskets of currencies.

2.9 Term of the Energy Storage Agreement

The term of the Energy Storage Agreement shall be **15 years**, commencing from the Commercial Operation Date.

2.10 Format, Sealing, Marking and Submission of Proposals

Two sealed packages containing;

1. The Original Proposal –
 - Clearly marked “Original” on the first page and last page.
 - An envelope marked “Proposal Security” contains the original of the “Proposal Security” shall carry the name and address of the Project Proponent and the Grid Substation should be included in the original proposal. The Proposal Security shall be substantially in accordance with the specimen given in Section 8 of Volume II.
 - In addition, a USB flash drive containing a scanned copy of the original document should be securely attached to the inside of the back cover of the Original Proposal document. The USB flash drive shall be clearly labelled with the name of the Project Proponent and the relevant Grid Substation.
2. The Copy –
 - Should contain a copy of the Proposal clearly marked “Copy” on the first page and last page.

Proposal shall be submitted separately and clearly labelled as “Original” and “Copy” including the Grid Substation. The cover of the two packages shall be labelled as below.

ORIGINAL / COPY

**PROPOSAL FOR BATTERY ENERGY STORAGE SYSTEM CONNECTED TO
.....GRID SUBSTATION
(CAPACITY: 10MW/40 MWh)**

PROJECT PROPONENT'S FULL NAME AND ADDRESS
.....

In the event of late delivery of the Proposal it shall be rejected and returned unopened to the Project Proponent whose address is stated on the package. **Under no circumstances proposals shall be accepted beyond the closing time.**

The Proposal containing all the above packages properly marked shall be delivered together in a sealed package to the following address on **July 24, 2026 at 10.00 hrs.**

The Chairman CANC
C/o; Director (RE & ESCP)
National System Operator (Pvt) Ltd.
Meethotamulla Road,
Kolonnawa 10600,
SRI LANKA

Note: This venue is subjected to change based on the responses to the tender.

If a proposal will be delivered **before July 24, 2026** , please handover **during office hours (8.30hrs - 16.00hrs)** to;

The Chairman CANC
C/o; Director (RE & ESCP)
National System Operator (Pvt) Ltd.
Meethotamulla Road,
Kolonnawa 10600,
SRI LANKA

TEL: +94 11 2697044 / +94 11 2583344

FAX: +94 11 2697022

2.11 Validity of the Proposal

Each Proposal shall constitute a firm offer and that shall remain in force and valid for a period of **One Hundred and Fifty (150) Calendar Days** following the Closing Date. During this period the proposed Bid shall remain valid and no Project Proponent shall withdraw its Proposal.

In the event of a Letter of Award not being issued during this period, NSO will notify each Project Proponent by fax **at least 14 days prior to the expiration date** requesting an extension of the Proposal validity period along with the extension of the validity of the Proposal Security. If any Project Proponent does not agree to such extension requested by NSO, **it may advise NSO of its decision in writing prior to the expiration of the original validity period of the proposal.**

If such decision is not received by NSO prior to the expiration date of the original validity period following the issuance by NSO of a notice of extension, the extension shall be considered to have been accepted by the Project Proponent, **provided that the Project Proponent shall extend the effective period of the Proposal Security** accordingly.

2.12 Closing Date and Late Proposals

Proposals received after 10.00hrs on the closing date, as stated in Clause 1.4 of Volume I of RFP, shall not be accepted nor considered whatever the reasons for delay in receipt (including circumstances outside the control of the Project Proponent).

2.13 Proposal Opening

Proposal and the envelope containing the Proposal Security will be opened immediately following the closing of bids by the relevant Tender Opening Committee specified for each grid substation as follows. Project Proponents or their authorised representatives who wish to attend the opening may do so **(only two participants per project proponent are allowed for each block)**.

Tender Opening Committee (Block)	Grid Substation
Tender Opening Committee 1	Ambalangoda
	Ampara
	Bolawatta
	Chunnakam
	Hambantota
Tender Opening Committee 2	Horana
	Killinochchi
	Kiribathkumbura
	Kosgama
	Kurunegala
Tender Opening Committee 3	Madampe
	Mahiyanganaya
	Matara
	Mathugama
	Monaragala
Tender Opening Committee 4	Galle
	Panadura
	Pannala
	Pannipitiya

Project Proponents' names, any withdrawals, quoted Capacity Charge Rate, the presence or absence of valid Proposal Security and such other details as NSO may consider appropriate will be announced at the time of opening of the Proposal. Information of which is of a commercially sensitive nature will not be disclosed.

2.14 Confidentiality

GOSL and NSO reserve the right to release information contained in the Proposals to its agents, consultants and advisers for purposes of verifying and evaluating the Proposals. Neither GOSL, NSO nor their agents, consultants and advisers shall be liable in any manner whatsoever for any loss or damages resulting from any disclosure of such information before, during or after the bidding process.

2.15 Clarifications & Correction of Errors

To assist in the examination, evaluation and comparison of Proposals, NSO may at any time prior to the final evaluation and issue of the Letter of Award request any Project Proponent for clarification of its Proposal, including breakdown of costs in determining the proposed Capacity Charge Rate.

Obvious arithmetical errors may be rectified. Where a Project Proponent wishes to correct other errors in relation to clarifications sought by NSO, which does not have an impact on the Capacity Charge Rate, NSO may accept such clarifications.

2.16 Cost of preparation of Detailed Proposal

The Project Proponents shall bear all costs and expenses associated with the preparation and submission of the Proposal. NSO shall not be responsible or liable for such costs and expenses, regardless of the conduct or outcome of the selection process. All costs and expenses involved in attending any meeting and/or visits to the site shall be at the sole cost and expense of the Project Proponent's.

2.17 Security Requirements

To guarantee the due performance of its obligations to construct and operate the Facility, the Project Proponent shall provide the following security instruments.

2.17.1 Proposal Security

Proposal security in the form of a Bank Guarantee issued by a commercial bank operating in Sri Lanka, approved by the Central Bank of Sri Lanka or a bank based in another country but the guarantee "confirmed" by a commercial bank operating in Sri Lanka and payable to the National System Operator, substantially in the form specified in Section 8, Volume II of the RFP.

Proposal shall be accompanied with a Proposal Security undertaking that the Proposal will be held valid for the period specified in Clause 2.11 of this document and that the Proposal shall not be withdrawn during that period. Proposal Security shall amount to **Sri Lankan Rupees Twenty Five Million (LKR 25,000,000/-) per 10MW/40MWh BESS Project**. Failure to submit the Proposal Security at the time or before the closing of the bidding and in accordance with above requirements will result in the Bid being rejected. No alternate offers will be accepted under one Proposal Security. The Proposal Security shall be unconditionally encashable, on the receipt of first written demand of the National System Operator and it should be irrevocable, unconditional and on-demand payable in Sri Lanka in Sri Lankan Rupees.

Failure to submit the proposal security as specified above is considered as a major deviation and the proposal will be considered as non - responsive.

The Proposal Securities of the Project Proponents **pending approval for awarding** will be returned after the expiry of the validity of the Proposal Security unless otherwise agreed to extend the same.

The Proposal Security shall be valid for no less than **one hundred & eighty (180) Calendar Days** from the Closing Date for submission of Proposal documents (valid until February 10, 2027). In the event the date for the issuance of the Letter of Award is extended by the NSO and the Project Proponent agrees to such extension, the Proposal Security shall be revalidated to cover such additional periods.

The NSO may forfeit the Proposal Security:

- a. If - Project Proponent withdraws its Proposal during the specified validity period; or
- b. In the case of a successful Project Proponent, if he fails
 - i) to furnish the Letter of acceptance within the period stated in Clause 6.6 of Volume I of RFP from the receipt of Letter of Award.
 - ii) to furnish the Performance Security within the period stated in Clause 2.17.2 of Volume I of RFP.

2.17.2 Performance Security

Within 30 days of receipt of the Letter of award or 7 days before the expiry of the Proposal Security, whichever comes first, the Successful Project Proponent shall furnish a Performance Security,

- In favour of the Project Proponent if the ESA is planned to be signed under Project Proponent's name
- In favour of the Special Purpose Company (SPV) along with relevant documents issued from Registrar of Companies (RoC) to substantiate 100% share requirement specified in Clause 5, Volume I of the RFP, if the ESA is planned to be signed in SPV name (*The performance security shall clearly indicate that this has been issued on the request of "Project Proponent" and in favour of "Special Purpose Company"*)

agreeing to execute the terms and conditions stated in Section 9, Volume II of the RFP.

The Proposal Security of a **successful Project Proponent** shall be discharged or returned, or both, (without interest) upon the Project Proponent's acceptance of the Letter of Award, and on furnishing of a Performance Security of **LKR Eighty Five Million (LKR 85,000,000/-) per 10MW/40MWh** valid for a period of 30 days beyond the commencement of Commercial Operations Date as defined in the Energy Storage Agreement.

This security shall be in the form of an irrevocable, unconditional and on demand Bank Guarantee issued by a reputed commercial bank operating in Sri Lanka or a bank based in another country but the guarantee "confirmed" by a bank operating in Sri Lanka, substantially in the form specified in Section 9, Volume II of the RFP.

The successful Project Proponent's Performance Security shall be discharged or returned, or both, (without interest) on Commercial Operation Date.

The NSO may forfeit the Performance Security:

- a. If the project proponent fails to sign the ESA, within the period specified in the Project

- b. If the project proponent fails to submit the deed/plan of the land as per Clause 6.6 of Volume I of RFP, within the period specified in the Project Milestone Schedule, Clause 1.4, Volume I of RFP for signing the ESA; or
- c. In case of Project Proponent fails to comply with the requirements of the Letter of Award issued to it; or
- d. The Project Proponent failure to achieve successful Commissioning & Testing by the date as specified in the ESA.

2.18 Project Proponent's Responsibilities

The submission of the Proposal shall be deemed to be acknowledged by the Project Proponent that it has carried out and performed the necessary inspections and investigations required to ascertain the suitability of the site and the local conditions, equipment to be furnished or installed, and other matters which may affect the performance of its obligations under the Project Documents.

The Project Proponent affirms by submitting its Proposal, that it has examined carefully and is fully familiar with all the work involved and this RFP shall become an integral part of the Project Documents and accepts without any reservation the terms and conditions specified in the RFP.

It also affirms that it has familiarised itself with the laws and regulations in force in Sri Lanka (including Labour, Financial, Environmental and Industrial laws) and that complying with these laws and regulations shall not affect the completion of the works under the Project Documents.

The failure or omission of a Project Proponent to obtain, receive or examine any form, document or to visit the site and acquaint itself with conditions existing there shall in no way relieve it from any obligation with regard to its Proposal.

3 PROJECT DESCRIPTION AND PERFORMANCE SPECIFICATIONS

The total installed plant capacity shall be as per Clause 1.3 of Volume I for each grid substation provided that it does not exceed the relevant grid substation capacity limits. The technology proposed for the project shall have a proven track record with demonstrated success in countries with a similar level of technological development and infrastructure support, as in Sri Lanka. Equipment offered shall be new and unused. All equipment and plant design must conform to NSO regulations applied to any power plant connected to Sri Lankan power system as stipulated in the Annex A, B and C, Volume I of the RFP. NSO has the right to refuse interconnection of any equipment that does not conform to this code and the Project Proponent must bear any loss incurred as a result.

3.1 Technical Information

Project Proponent shall provide general technical information, where possible with Technical Literature to enable NSO to understand the operation of the plant and its peripherals / auxiliaries. The following minimum data shall be provided in this regard.

- a. Technical Details for BESS System as Section 6 of Volume II of RFP.
- b. BESS Modules to be installed should comply with the relevant International Standard as per Section 6 of Volume II of RFP (IEC 62619, IEC 62620, IEC 62902, IEC 62485-5, IEC 63056, IEC 62933-1, IEC 62933-2-1, IEC 62933-5-2, UL 9540, UL 9540A, UL 1973, UN 38.3, etc or latest available equivalent Standards)

Inverters and Power Conditioning Systems shall comply with the latest editions of the following standards: IEEE 1547-2018 (Standard for Interconnection and Interoperability of Distributed Energy Resources), IEEE 2800-2022 (Standard for Interconnection and Interoperability of Inverter-Based Resources Interconnecting with Associated Transmission Electric Power Systems), UL 1741-SB (Inverters, Converters, Controllers and Interconnection System Equipment, Supplement B for Grid Support Utility Interactive Inverters), IEC 62477-1, IEC 62109-1/2, IEC TS 62786-3 (stationary battery energy storage connected to the grid), and the relevant requirements of the Sri Lanka Grid Connection Code dated 22 July 2024

(Documentary proofs should be provided along with the Proposal)

- c. The BESS is required to operate in true grid-forming (V/F) mode. Grid forming is as follows, and as required to meet the grid connection specifications:
 - The BESS must have a full grid forming capable inverter (i.e. voltage sourced - voltage controlled operation) irrespective of other generation in the grid. It should not change the control mode to current controlled (i.e. grid following) during the normal operation or under any network faults conditions. This will be the main mode of operation for this BESS facility. Contactor shall provide sufficient documentary and modelling evidence to support the above requirement.
 - BESS must be capable of operating in parallel with other thermal generation, renewable energy generation, and BESS systems also operating in a voltage controlled mode in the grid.
 - BESS must be capable of receiving real power, reactive power, power factor and voltage set points or targets for charging/discharging or reactive power control while in grid forming mode depending on the excess solar PV generation or load balance in the grid.

- d. Transformers and switchgear details, demonstrating conformance with Section 6 of Volume II of RFP.
- e. Plant layout and Single line diagram up to the Grid Point showing all circuit breakers, isolators, current & potential transformers for metering and protection of the Facility, earthing switches, lightning arrestors, power transformers and inverters and dynamic model of the plant.
 - Single Line Diagram shall clearly specify the method adopted to limit the maximum power output (10MW+10%) at the termination point. Supporting documents to verify the same shall be submitted with the proposal.
- f. Expression of Interest from each of the prospective BESS (Battery Energy Storage System) suppliers to supply the required number of battery modules, inverters, and associated control systems for the project, along with relevant literature, brochures, or technical documents describing the manufacturer's business, facilities, and organizational capabilities.
- g. The Project Proponent shall ensure that the proposed Facility is designed, constructed, operated, and maintained in accordance with applicable international standards and industry best practices for fire prevention, fire detection, fire suppression, and emergency response systems relevant to Battery Energy Storage Systems (BESS). The proposed fire protection arrangements shall include, but not be limited to, fire detection systems, alarm systems, fire suppression systems, thermal runaway mitigation measures, fire compartmentalization, emergency isolation facilities, and appropriate safety procedures to minimize risks to personnel, equipment, and surrounding infrastructure. Documentary evidence demonstrating compliance with the applicable standards and fire safety requirements shall be submitted together with the Proposal.
- h. Type & ratings of major equipment, MV & LV switch gear, etc. (For this purpose, the maximum fault levels for each grid substation are given below;

	Grid Substation	Maximum 3-Phase Fault Level at 33kV level (kA)
1	Ambalangoda	11.2
2	Ampara	10.2
3	Bolawatta	11.2
4	Chunnakam	7.8
5	Hambantota	11.6
6	Horana	14.4
7	Killinochchi	7.9
8	Kiribathkumbura	16.2
9	Kosgama	9.0
10	Kurunegala	10.2
11	Madampe	13.7
12	Mahiyanganaya	9.0
13	Matara	12.7
14	Mathugama	13.0
15	Monaragala	4.5
16	Galle	11.6
17	Panadura	13.4
18	Pannala	10.7
19	Pannipitiya	14.4

Protection scheme proposed to be provided for the bus-bars, transformers, inverters and other applicable switchgears, including fire protection to comply with the provisions of Grid Connection Code and “CEB Guide for Grid Interconnection of Embedded Generators” as amended. Protection schemes may have interfacing with power system equipment and these protective schemes shall be subjected to the concurrence of NSO.

- i. Details of the metering devices at the Facility including Grid Code compliance monitoring system, power and energy meters and recorders. Please note that these meters/recorders shall meter, record, and transmit the data on the power/energy flow to the NSO system.
- j. Any other detail, the Project Proponent deems necessary.
- k. The storage capacity of the BESS, expressed in megawatt-hours (MWh), shall be directly proportional to the quoted rated power output, (in MW). The offered BESS shall be capable of continuously discharging at the quoted rated power output for a minimum duration of four (4) hours under ambient site conditions at the time of commissioning. Any proposal that does not meet these requirements shall be deemed non-compliant and will not be considered for evaluation.
- l. All non-battery equipment shall be designed for a minimum service life of 20 years under site conditions. Battery cells, modules and racks shall have a design life of at least 15 years when operated within the duty cycle specified in the ESA

- m. a Capacity Maintenance Plan describing how the Minimum Dispatchable Capacity schedule will be met across the 15-year term. The Plan shall include: (a) initial sizing strategy (Beginning-of-Life energy at the Termination Point versus contracted Energy Rating); (b) augmentation schedule with anticipated capacity additions in MWh and timing; (c) module/rack replacement intervals; (d) interface management between the Project Company and NSO during augmentation works including outage planning; (e) end-of-life decommissioning and battery recycling commitments compatible with applicable Sri Lankan environmental regulations on Environmental and Social Sustainability, which shall apply throughout the operating period and at end-of-life disposal.
- n. Project proponent shall report, on a monthly basis, actual dispatched energy, achieved RtE, system availability, and battery state-of-health (% nameplate) for each independently controllable BESS unit.
- o. It shall be mandatory to provide grid support services in accordance with A.05 of Annex A – Functional Performance Requirements – Grid Integration. The Project Proponent shall submit, on a monthly basis, a detailed report derived from the EMS event records demonstrating the grid support services provided by the BESS
- p. Submit initial non-site-specific simulation models of the BESS in both PSS®E and PSCAD™/EMTDC™ formats along with the tender for evaluation; or provide test results with the tender, demonstrating the BESS response in terms of voltage (V), active power (P), and reactive power (Q) during deep and shallow faults, and for a +50-degree phase angle step change at the Point of Connection (POC), under Short Circuit Ratio (SCR) levels of 1, 3, 5, and 10, with an X/R ratio of 5. Failure to meet at least one of the above requirements may result in the rejection of the bidder's technical proposal.

3.2 Power Transmission

At the **Metering Point** EDL fixes its metering equipment for the measurement of energy output and the title of electrical energy passes to EDL.

At the **Grid Point** the power transmission line/s physically connect to the national grid. From Termination Point to the Grid Point, the developing cost of power transmission line/s and all associated equipment shall be borne by the project company (as per the general arrangement proposed in Section 13 of Volume II of RFP) and owned/maintained by EDL.

The Project shall be designed for interconnection with the transmission and/or distribution network of the Transmission Licensee, as applicable, in accordance with the provisions of the Sri Lanka Electricity Act, 2024, as amended from time to time, the applicable Grid Code, Distribution Code, Grid Interconnection Standards, metering requirements, communication protocols, and all other rules, regulations, guidelines, and procedures issued by the Public Utilities Commission of Sri Lanka (PUCSL), the National System Operator, and the Transmission Licensee.

The Battery Energy Storage System (BESS) Facility shall be connected to the Medium Voltage (MV) distribution network with the coordination of respective Distribution Licensee, and the Developer shall be responsible for designing, engineering, procuring, constructing, testing, commissioning, operating, and maintaining all MV interconnection facilities required for evacuation of power from the BESS Facility up to the Point of Interconnection.

The Point of Interconnection shall be established at the Grid Substation ("GSS") premises, where the existing feeder shall be split through a gantry arrangement to provide a dedicated circuit for the BESS Facility. The revenue metering point for energy accounting shall be located at the said gantry. Developer shall establish and maintain a dedicated MV feeder from the BESS Facility up to the Point of Interconnection. All associated works, including line routing, poles/cables, switchgear, protection systems, SCADA/communication interfaces, and any required augmentation of the MV network to ensure compliance with grid requirements, shall be the sole responsibility of the Developer and shall be undertaken entirely at its own cost and expense, subject to approvals of the Transmission Licensee.

All losses and auxiliary consumption occurring in the dedicated MV interconnection system between the BESS Facility and the revenue metering point at the GSS shall be borne by the Developer.

Refer Section 13 & 14 Volume II of RFP for further details.

3.3 Telecommunications equipment for communications, telemetry and control

Project company shall provide Industrial grade, independent SCADA gateway for integration with National System Control Centre. The above gateway shall comply with the following specifications;

- SCADA gateways shall comply with the IEC 60870-5-104 for Master station standards.
- Equipment and port redundancy shall be available for the SCADA gateway. In case of a main one failure, standby unit shall be able to switch to main operation with NSCC.
- The developer shall provide a reliable technical solution to integrate the data feeds from dual redundant sources to the gateways, ensuring the seamless SCADA reporting with precise time stamps.

All the signals related to BESS Facility as per typical signal lists attached as Annex D of Volume I of RFP shall be configured in the Gateway as per the IEC 60870 - 5 - 104 protocol and made available to SCC by Project Company via the NTNISP communication equipment at the relevant Grid Substation before the commissioning. Gateway configuration parameters are given in Annex D of Volume I of RFP. The communication interfaces shall be configured as in Annex D of Volume I of RFP. Finalized Signal List shall be submitted at least 02 weeks prior to commissioning.

NSO shall provide communications channels via fiber optic (FO) cable from Grid Substation to link the BESS with the NTNISP Communication network for provision of Voice and SCADA

a. Fiber Optic Cable from BESS to Grid

Two (02) nos. separate fiber optic cables (for redundancy) shall be drawn from BESS to the Grid Substation.

The fiber cable shall contain 8 nos. of single mode fiber cores suitable for transmission of 1310 nm and 1550nm optical wavelengths and they shall be in conformity with ITU-T recommendations G.652.D.

A suitable distribution frame/patch panels with single mode pig tail (FC) terminations shall be installed at two ends of FO cables to terminate 48 Nos. of cores. Suitable all dielectric ruggedized optical fiber approach cables shall be used from the outside switchyard joint box to the indoor patch panel of Grid Substation.

- b. A suitable DC powered firewall with redundant power supply units shall be supplied, installed and commissioned (to suit the NSO/NSCC recommended security rules/policies) at the Grid Substation. And from this firewall there shall be 02 Nos. of parallel 1 Gbps optical Ethernet links on the two redundant optical fiber cables, commissioned up to an appropriate firewall/network switch installed at the BESS end. The firewall at Grid Substation may require NAT (network Address translation) facility for masking the internal subnets.
- c. At the grid substation from this new firewall there shall be 02 Nos. of Electrical 1000BaseT type parallel links (trunks) to the existing optical multiplexer to interface SCADA and Voice over IP facilities tagged with IEEE 802.1Q VLANs.
- d. An IP desktop phone shall be made available at BESS to provide operational voice communication with NSCC.
- e. The specification of the IP phone and the firewall shall be sent to NSO for reviewing and necessary approvals.

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4 ENVIRONMENTAL CONSIDERATIONS

Environmental Requirements

Within the scope of the project, the Project Proponent is required to assess the environmental concerns of the proposed Battery Energy Storage System and obtain all relevant environmental clearances and approvals from the appropriate authorities in terms of applicable laws and regulations and the guidelines published by Central Environmental Authority.

Project Proponent is required to comply with National Environment Act No. 47 of 1980, National Environment (Protection & Quality) Regulations, No 1 of 1990, and to any amendments thereto, and also to any other applicable regulation or law. The Project Company shall comply with all Environmental Laws applicable to the Facility during the Term.

4.1 Archaeological Requirements

In addition to the above requirements, the Project Proponents are required to comply with the requirements of the Antiquities Ordinance, as amended by Act No. 24, 1998 and regulations there under.

5 DETAILS OF THE PROJECT PROPONENT

Project Proponents who wish to submit proposals may do so either as, an incorporated new company, or as an incorporated existing company or as a joint venture/consortium of such companies.

Eligible Bidders:

All foreign and local prospective Project Proponents are eligible.

However, it should be noted:

- a. Project Proponent whose Proposal has been declared as successful shall not dispose of their interest or withdraw in relation to the proposal prior to execution of the Project Agreements referred in Clause 7, Volume I of RFP ("Project Agreements").

The Project may be undertaken by a special purpose company incorporated in Sri Lanka with limited liability. 100% shares of the Special Purpose Company are from the project proponent company.

Project Proponents shall also be required to meet the following criteria:

- a. No Project Proponent or its parent/affiliate companies shall have been suspended or black listed by NSO, the GOSL or by any other Government Agency in Sri Lanka, whether as an individual contractor or corporation or as a member of a joint venture/consortium.
- b. No Project Proponent or its parent/affiliate companies shall have a record of unsatisfactory past performance, particularly non-compliance with contractual terms, plans and specifications; nor any record of defective workmanship and materials supplied, abandonment of work or similar efficiencies.
- c. No Project Proponent or its parent/affiliate companies shall have inordinate overdue debts toward the Insurance, Revenue or Customs authorities of Sri Lanka or in any other country.
- d. No Project Proponent or its parent/affiliate companies shall have criminal or civil cases in court pending or finally decided against them involving non-payment of tax, duty, statutory dues or other undertaking with any Government or its sub-divisions, or instrumentalities including government-owned or controlled corporations.

6 EVALUATION AND ISSUANCE OF THE FORMAL INVITATION

6.1 Outline of Evaluation Procedure

The actual evaluation process is a staged process. In the first stage, the responsiveness of the Proposals will be assessed for the compliance with the requirements and the procedures laid out in this RFP. The second stage of the evaluation process concentrates on the requirement of technical and commercial details of the Proposal. In the third stage, the evaluation will be based on the proposed Capacity Charge Rate as specified in Clause 6.5 of Volume I.

6.2 Right to Reject Proposals

NSO reserves the right to reject the Proposal of any Project Proponent who has qualified on the basis of fraudulent, suppressed or incomplete information. Project Proponents who submit proposals do so without recourse against GOSL, or NSO or their representative for either rejection by NSO or failure to execute a ESA, for any reason whatsoever.

6.3 First Stage Evaluation (Responsiveness)

In order to facilitate and expedite the evaluation process, the Project Committee shall initially evaluate only the lowest-priced Proposals received at the time of the closing of submission of Proposals. For all Grid Substations, the lowest-priced three (3) Proposals shall initially be considered for evaluation. However, for the following Grid Substations, the lowest-priced four (4) Proposals shall initially be considered for evaluation:

- Mathugama
- Kurunegala
- Bolawatta
- Matara
- Galle
- Chunnakam

The Project Committee reserves the right to evaluate additional Proposals, if required, in the event that any of the initially evaluated Proposals are determined to be non-responsive or fail to satisfy the requirements specified in this RFP.

NSO will examine the Proposals to determine whether they are complete, whether the required Proposal Security has been furnished, whether the documents have been properly signed, whether all annexes as required in the RFP are furnished and whether the Proposals are generally in order. A Proposal may be disqualified and excluded from further consideration for any valid reason including but not limited to, the reasons listed below:

- a. Failure to submit supporting documentation or any other clarification or any documents requested by NSO within the required time frame.
- b. Failure to submit **Original of the Certificate of Purchase of RFP** issued by the NSO
- c. Failure to submit Duly executed Power of Attorney or Board Resolution to the authorized representative who is to sign on behalf of the Project Proponent.
- d. Failure to submit Document/s to confirm the ownership or the ability to procure or lease the

selected land as per Clause 1.3.1 of RFP Volume I.

- e. Failure to complete any of the Proposal Forms.
- f. Willful misrepresentations in the Proposal.
- g. Illegal conduct or attempts to influence the GOSL, NSO, Evaluation Committee, Procurement Committee in evaluation of a Proposal outside the solicitation process.

6.3.1 Major Deviation/s.

Failure to submit any of the following items with the proposal will be considered as a Major Deviation and the proposal **will not be considered for further evaluation**.

1. Duly completed Proposal Letter signed by the authorized person
2. Proposal Security,
 - Substantially in the given format (Section 8, Volume II of RFP) without any conditions or
 - for the specified validity period or
 - for the specified value or
 - issued by a reputed commercial bank operating in Sri Lanka or by a bank based in another country with the guarantee "confirmed" by a commercial bank operating in Sri Lanka or
 - issued in the name of the Project Proponent.
3. Certification of Registration under the Public Contract Act No. 3 of 1987 -PCA3.
4. Documentary evidence demonstrating compliance with the minimum manufacturer qualification requirements specified in Clause 2.7.3 of Volume I of the RFP.
5. Documentary proof demonstrating the availability of adequate finances, or sufficient details on the manner in which the required finances for the Project are proposed to be secured, in accordance with the requirements specified in this RFP.
6. Duly completed Capacity Charge Rate Proposal signed by the authorized person (Section 4, Volume II of RFP)

If a Proposal is found to be non-responsive, it will be rejected, and may not subsequently be made responsive by correction or withdrawal of the item which resulted in the disqualification and/or rejection.

6.4 Second Stage Evaluation

The evaluation will be based on the information provided in the Proposal. The Proposals will be evaluated on the following criteria:

- | | |
|----|---------------------------------------|
| 1. | Finance ability of the Project |
|----|---------------------------------------|

1.1	Supporting documents to verify the proposed debt components from the bank as per Clause 2.8.2, Volume I of RFP
1.2	Supporting documents to verify the proposed equity components of individuals or joint venture/consortiums as per Clause 2.8.2, Volume I of RFP
2. Specification of the plant	
2.1	Technical Information as per Clause 3, Volume I of the RFP.

Non-compliance with any one of the above criteria may disqualify the Project Proponent.

6.5 Third Stage Evaluation

The responsive Project Proponents of the above stages shall thereafter be ranked in ascending order of the Capacity Charge Rate (lowest to highest).

The successful Project Proponents for each grid substation will be chosen on the basis of the above ranking.

If a successful project proponent fails to submit the Letter of Acceptance within the time frame specified in Clause 1.4, Volume I of the RFP, the next unselected responsive lowest bidders in the same grid substation will be invited. However, the invited Project Proponents have to adjust the proposed Capacity Charge Rate to the Capacity Charge Rate of the withdrawn bid.

The successful Project Proponents will be issued formal Letters of award. Following submission of the Letters of Acceptance along with the Performance Security as specified in Clause 2.17.2, Energy Storage Agreements will be entered into for the bid values as stipulated in Clause 7 of RFP volume I, after the bidding process on variable price option is completed. However, nothing mentioned herein shall prejudice the right of NSO to terminate this process and/or refrain from issuing the formal Letter of Award as mentioned above.

If the project proponents having the same rank (due to offering same Capacity Charge Rate) affects the selection for a specific grid substation, following procedure will be followed.

1. Each such Project Proponent will be invited to submit three alternate sealed Capacity Charge Rate (proposals marked as 1,2,3 each containing a Capacity Charge Rate lower than the previously quoted Capacity Charge Rate.
2. Envelops 1, 2, 3 should contain Capacity Charge Rate in descending order. For example, the price quoted in envelop 2 should be lower than the price in envelop 1 and the price quoted in envelop 3 should be lower than price in envelop 2.
3. Such Project Proponents will be called upon to a special proposal opening meeting and envelop 1 of all such Project Proponents will be opened first.
4. Based on the prices quoted in envelop 1, the lowest evaluated Proposal/s will be selected.
5. If the issue of project proponents having the same rank (due to offering same Capacity Charge Rate) still remains, envelop 2 of relevant project proponents will be opened.

6. This will be continued if required, until all three envelopes are opened.
7. All unopened envelopes, if any, will be returned.

During this stage if any Project Proponent does not agree to provide a lower Capacity Charge Rate than his originally quoted price as requested above, he may officially communicate his intention to NSO and withdraw his Proposal from the bidding process, **without forfeiting his Proposal Security**.

NSO reserves the right to terminate the RFP process if adequate competition has not been created or and the Capacity Charge Rates quoted by project proponents are not acceptable to the NSO.

6.6 Letter of Award

The successful Project Proponents will be issued formal Letter of Award to submit their Letter of Acceptance within 7 days of the receipt of the Letter of Award.

In order to issue proceed to next steps for signing Energy Storage Agreement (ESA) it is mandatory to submit the following documents to NSO;

1. Acceptance of Letter of Award within 7 days of the receipt of the Letter of Award
2. Performance Security with in the period specified in 2.17.2, Volume I of RFP
3. Duly certified copies of documents below within **30 days** from the date of receipt of the Letter of Award
 - 3.1 Certificate of Incorporation
 - 3.2 List of the Board of Directors
 - 3.3 Location of the Registered Office
 - 3.4 Resolution of the Board of Directors authorizing the persons entitled to witness the affixing of the Company Seal
 - 3.5 Land Details;

For private land:

 - A copy of the Deed of the land in the name of the project proponent certified by an Attorney-at-Law with the survey plan
 - A copy of the Lease Agreement (at least for 17 years) of the land in the name of the project proponent certified by an Attorney-at-Law with the survey plan

For LRC land / Land owned by state institutions / State land

 - A confirmation letter in the name of the project proponent and issued by the relevant authorised officer as specified in Clause 1.3.1.
 - 3.6 Map (Colour, A3) indicating the plant location, address and GPS coordinates

Non-submission of the above within the period specified above will result in forfeiture of the Proposal/Performance security respectively and the ESA will not be signed.

If a successful Project Proponent fails to submit the Letter of Acceptance within the time frame specified in the RFP, the NSO in its discretion may invite the next ranked Project Proponent for the relevant grid substation as per conditions in Clause 6.5 of Volume I of RFP.

The Proposal Securities of the **unsuccessful Project Proponents** will be returned upon the receipt of Letter of Acceptance from the successful Project Proponents or the expiry of the validity of the Proposal Securities unless or otherwise agreed to extend the same.

The Project Proponent shall submit, within two (2) months from the acceptance of the Letter of Award, all results and related documentation pertaining to the Dynamic Model Tests as stipulated in the Functional Performance Requirements and Grid Connection Code Requirements (Annex A, Annex B). Construction of Facility shall commence only upon receiving written concurrence from NSO on the submitted Dynamic Model Test results.

6.7 Cost and Risk in Finalising Project Agreements

The successful Project Proponents shall bear all costs incurred by it in relation to the finalization of the Project Agreements, including but not limited to those of professional advisers. Neither GOSL, NSO, or any representative of these parties shall have any liability whatsoever to the Project Proponent for any decision taken by the Project Proponent in relation to decisions taken by it in finalising and entering into the Project Agreements, whether or not in reliance on any matter supplied or represented by GOSL, NSO, or their representatives.

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7 PROJECT AGREEMENTS

7.1 Introduction

Project Agreement is the model Energy Storage Agreement (ESA) set out in Volume III of this RFP. The Project Proponents should study the Project Agreement in detail when formulating their Proposals.

The terms and conditions contained in the Energy Storage Agreement (ESA) shall be deemed to have been accepted by the Project Proponent upon submission of the Proposal. Following the issuance and acceptance of the Letter of Award, no qualifications, reservations, proposed amendments, deviations, or modifications to the terms and conditions of the ESA shall be entertained. The successful Project Proponent shall execute the ESA in the form provided in Volume III of this RFP without alteration subject to any revisions recommended by the Attorney General.

7.2 Requirement to Sign Energy Storage Agreement

As per the model Energy Storage agreement (Volume III of the RFP), the Project should be a “company duly incorporated and validly existing under the laws of the Democratic Socialist Republic of Sri Lanka, has all requisite corporate and legal authority to execute this Agreement, and is permitted by applicable laws and regulations to sell independently produced power”.

7.3 The Energy Storage Agreement

The Energy Storage Agreement is signed between NSO and the Project Company and relates to the storage services provided by the facility. The Energy Storage Agreement will be signed only after obtaining all other statutory clearances. A prescribed ESA processing fee is applicable.

8 ANNEXES

Annex A: Functional Performance Requirements – Grid Integration

Annex B: Grid Connection Code (July 2024)

Annex C: CEB Guide for Grid Interconnection of Embedded Generators

Annex D: Signal List for SCADA integration with National System Control Centre

Note:

Some clause specifically mentioned in Functional Performance Requirements issued with the RFP, may differ from the Grid connection Code. In such instance the Functional Performance Requirements issued with the RFP shall take precedence over the Grid Connection Code requirements.

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